

5G RAN

Multi-Band Compatibility Feature Parameter Description

Issue	Draft A
Date	2025-12-31



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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue does not introduce any changes to 5G RAN9.1 02 (2025-08-30).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FBFD-010014	Mobility Management	4 Multi-Band Compatibility

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Multi-Band Compatibility	Supported in both NSA and SA networking, with the following differences: In NSA networking, only the following functions are supported: handover to a multi-band cell, carrier aggregation, CoMP, and neighboring cell configuration. Selection and reselection to an intra-NR multi-band cell are not supported. In SA networking, all functions are supported. Neighboring cell configuration is supported with slight differences. For details, see 4.1.4.1 Neighboring Cell Configuration .	4 Multi-Band Compatibility

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Multi-Band Compatibility	Supported in both high and low frequency bands, with the following differences: SA networking is supported only in FWA scenarios in high frequency bands and is supported in all scenarios in low frequency bands.	4 Multi-Band Compatibility

3 Introduction

3GPP specifications define the mapping between NR frequency bands and frequencies. Some frequencies map to more than one frequency band. For example:

Frequencies in the range of 3300 MHz to 3800 MHz are included in bands n77 and n78.

Frequencies in the range of 26500 MHz to 27500 MHz are included in bands n257 and n258, and frequencies in the range of 27500 MHz to 28350 MHz are included in bands n257 and n261.

For details, see section 5.2 "Operating bands" in 3GPP TS 38.101-1 V15.6.0.

If the operating frequency of a cell maps to more than one frequency band, the cell is considered a multi-band cell. After Multi-Band Compatibility is introduced, a cell is considered available once a UE supports any of the cell's frequency bands. This allows UEs with a wider variety of bands to access the cell, thereby increasing the number of online UEs.

4 Multi-Band Compatibility

4.1 Principles

4.1.1 Selection to a Multi-Band Cell

A multi-band cell can be configured with up to two frequency bands: one primary and one additional frequency band, which are specified by the **NRDUCell.FrequencyBand** and **NRDUCell.AdditionalFrequencyBand** parameters, respectively. The settings of these parameters are delivered to a UE through the *frequencyBandList* IE in System Information Block 1 (SIB1), allowing the UE to select a matching frequency band.

A UE in idle mode selects a multi-band cell to camp on according to the following logic:

- If the UE only supports the primary frequency band, the UE camps on this band.
- If the UE only supports the additional frequency band, the UE camps on this band.
- If the UE supports both the primary and additional frequency bands, the UE camps on the primary frequency band.

 NOTE

Selection to an intra-NR multi-band cell is not involved in NSA networking.

4.1.2 Reselection to a Multi-Band Cell

A UE in idle mode can reselect to a neighboring multi-band NR cell. During cell reselection, the UE measures the neighboring frequencies included in the system information broadcast by the gNodeB. If a neighboring cell is a multi-band cell, the system information contains a primary and an additional frequency band. The primary and additional frequency bands are specified by the **NRCellFreqRelation.FrequencyBand** and **NRCellFreqRelation.AdditionalFrequencyBand** parameters, respectively.

SIB4 carries information about neighboring NR frequencies, and broadcasts cell reselection parameters which match those in the **NRCellFreqRelation** MO. A UE

performs cell reselection based on its band capability. For details about cell reselection, see *SA Mobility Management in Idle Mode*.

 NOTE

Reselection to an intra-NR multi-band cell is not involved in NSA networking.

4.1.3 Handover to a Multi-Band Cell

A UE in connected mode can be handed over to a neighboring multi-band NR cell. During a handover, the gNodeB delivers neighboring frequencies for the UE to measure. If the neighboring cell is a multi-band cell, the principles for the delivery of these frequencies and handover decision-making are as follows:

Delivery of Frequencies to Be Measured

The gNodeB delivers information about effective frequencies to the UE for measurement according to the supported frequency bands reported by the UE and the configured cell frequencies.

- If the UE supports only the primary frequency band, information specific to the primary frequency band is delivered.
- If the UE supports only the additional frequency band, information specific to the additional frequency band is delivered.
- If the UE supports both the primary and additional frequency bands, information specific to the primary frequency band is delivered preferentially.

 NOTE

If the *bandNR* IE in **supportedBandCombinationList** of the **rf-Parameters** field in the **UECapabilityInformation** message over the Uu interface indicates that a UE does not support the primary frequency band or the additional frequency band, the gNodeB does not deliver event A3 measurement configurations to the UE after it accesses the gNodeB. For details about UE capabilities, see "RF-Parameters" in section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V18.4.0. For details about event A3 measurement configurations, see *SA Mobility Management in Connected Mode*.

Decision-Making in Handovers

If a UE supports either frequency band of the target cell, handovers are allowed. Otherwise, handovers cannot be performed.

For details about handovers, see *SA Mobility Management in Connected Mode*.

 NOTE

The PSCell change procedure in NSA networking is the same as that for a handover to a multi-band cell in SA networking. For details about the PSCell addition procedure in NSA networking, see *EPC-based NSA Basics*.

4.1.4 Implementation of Other Functions in a Multi-Band Cell

4.1.4.1 Neighboring Cell Configuration

The gNodeB updates the frequency bands of an external neighboring NR cell based on the following:

- UE-reported measurement results
The gNodeB generates the frequency band information of the external neighboring NR cell based on the *FrequencyBandList* IE in CGI-Info measured and reported by the UE. The first frequency band in CGI-Info is updated to **NRExternalNCell.FrequencyBand**, and the other frequency band is updated to **NRExternalNCell.AdditionalFrequencyBand**.
- Xn messages
When an Xn interface is set up or updated, the gNodeB updates the frequency band information of the external neighboring NR cell based on the *NR Frequency Band Item* IE contained in the Xn message.
 - If the Xn message only contains information about the primary frequency band, such information is updated to **NRExternalNCell.FrequencyBand**.
 - If the Xn message contains information about more than one frequency band, the primary frequency band is updated to **NRExternalNCell.FrequencyBand**, and the other frequency band is updated to **NRExternalNCell.AdditionalFrequencyBand**.

 NOTE

- When the update is based on the UE-reported measurement results in NSA networking, neighboring cells need to send SIB1 for UEs to obtain CGI-Info. The gNodeB then generates frequency band information of external neighboring NR cells.
- The update based on Xn messages is not involved in NSA networking.

4.1.4.2 Carrier Aggregation

In carrier aggregation (CA) scenarios, to determine whether an aggregation cell that is a multi-band cell can be selected for CA, the gNodeB compares the primary and additional frequency bands of the cell with the CA band combinations supported by a CA UE. If an operating band of the cell is included in the CA band combinations supported by the UE, the cell can be configured as a Secondary Cell (SCell) for the CA UE.

For details about CA, see *Carrier Aggregation*.

4.1.4.3 CoMP

In CoMP scenarios, the gNodeB first checks whether the target cell and serving cell have the same primary frequency band before selecting a coordinating cell. Then, the gNodeB compares the UE's band capability with the primary and additional frequency bands of the target cell. The cell can be configured as a coordinating cell once the UE's frequency band matches either of its frequency bands. For details about CoMP, see *CoMP*.

4.2 Network Analysis

4.2.1 Benefits

After Multi-Band Compatibility is enabled, a UE can access a multi-band cell if it supports any of the frequency bands of the cell. This allows UEs with a wider

variety of frequency bands to access the cell, thereby increasing the number of online UEs, and facilitating UE management and operation.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

- On existing networks, set the current frequency band as the primary frequency band and the other frequency band as the additional frequency band. For new networks, it is recommended that the frequency band supported by the most UEs be configured as the primary frequency band.
- The settings of the primary and additional frequency bands must be consistent network-wide for multi-band cells that operate on the same frequency.
- The primary frequency band of a multi-band cell must be the same as that of a common intra-frequency cell.
- The primary and additional frequency bands configured in the **NRCellFreqRelation** and **NRExternalNCell** MOs must be the same.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Before activating this function, configure the NR DU cell, frequency relationship of the NR cell, and frequency bands of external neighboring NR cells. During configuration, ensure that the primary and additional frequency bands configured in the **NRCellFreqRelation** and **NRExternalNCell** MOs are the same. [Table 4-1](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

NOTE

In NSA networking, multi-band configurations on the LTE side must be performed. For details, see *EPC-based NSA Basics*.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Frequency Band	NRDUCell.FrequencyBand	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Additional Frequency Band	NRDUCell.AdditionalFrequencyBand	Set this parameter based on the network plan.
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.
Additional Frequency Band	NRCellFreqRelation.AdditionalFrequencyBand	Set this parameter based on the network plan.
Frequency Band	NRExternalNCell.FrequencyBand	Set this parameter based on the network plan.
Additional Frequency Band	NRExternalNCell.AdditionalFrequencyBand	Set this parameter based on the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples (TDD)

Configure an NR DU cell as a multi-band cell that, for example, supports both n77 (primary frequency band) and n78 (additional frequency band).

```
//Configuring an NR DU cell
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N77, Dlnarfcn=634000, UlBandwidth=CELL_BW_100M,
DlBandwidth=CELL_BW_100M, SlotAssignment=4_1_DDDSU, SlotStructure=SS2, TrackingAreaId=0,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=7853, LogicalRootSequenceIndex=80,
AdditionalFrequencyBand=N78;
```

Configure the frequency relationship of the NR DU cell, with its neighboring cell configured as a multi-band cell that, for example, supports both n77 (primary frequency band) and n78 (additional frequency band).

```
//Configuring an NR cell frequency relationship
ADD NRCELLFREQLATION: NrCellId=0, SsbFreqPos=7800, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, AdditionalFrequencyBand=N78;
```

Configure the external neighboring NR cell as a multi-band cell that, for example, supports both n77 (primary frequency band) and n78 (additional frequency band).

```
//Configuring an external neighboring NR cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, PhysicalCellId=0, Tac=0,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=7800, FrequencyBand=N77,
AdditionalFrequencyBand=N78;
```


Optimization Command Examples

Function optimization is not involved, and therefore no parameters are required for function optimization.

Deactivation Command Examples (TDD)

```
//Modifying the NR DU cell with AdditionalFrequencyBand set to NULL  
MOD NRDUCELL: NrDuCellId=0, AdditionalFrequencyBand=NULL;  
//Modifying the frequency relationship of the NR cell with AdditionalFrequencyBand set to NULL  
MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=7800, AdditionalFrequencyBand=NULL;  
//Modifying the external neighboring NR cell with AdditionalFrequencyBand set to NULL  
MOD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, AdditionalFrequencyBand=NULL;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Frequency Bands of the NR DU Cell

Step 1 Log in to the MAE-Access, and choose **Monitor > Signaling Trace > Signaling Trace Management > TraceType > NR > Application Layer > Uu Interface Trace**.

Step 2 Check the frequency band information in SIB1. The first and second **FreqBandIndicator** values indicate the primary and additional frequency bands, respectively.

If the frequency band information in SIB1 is consistent with that in the cell configuration, the configuration is correct.

Step 3 Run the **DSP NRCELL** command to check the cell status.

If the value of **Cell Available State** in the command output is **Available**, the cell has been set up successfully.

Step 4 Verify cell functions.

If a UE that works only in the additional frequency band successfully accesses the cell, the cell is functioning properly.

----End

Frequency Bands Configured in the NR Cell Frequency Relationship

Step 1 Log in to the MAE-Access, and choose **Monitor > Signaling Trace > Signaling Trace Management > TraceType > NR > Application Layer > Uu Interface Trace**.

Step 2 Check the frequency band information in SIB4. The first and second **FreqBandIndicator** values indicate the primary and additional frequency bands, respectively.

If the frequency band information in SIB4 is consistent with that in the NR cell frequency relationship, the configuration is correct.

----End

Frequency Bands of the External Neighboring NR Cell

Run the **LST NREXTERNALNCELL** command to display the frequency bands of the external neighboring NR cell. If the values of **Frequency Band** and **Additional Frequency Band** are the same as those of the NR DU cell of the corresponding gNodeB, the band configuration is correct.

4.4.3 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Idle Mode*
- *Carrier Aggregation*
- *CoMP*
- *EPC-based NSA Basics*